

Arrow FPGA Starter Kit – AXE5000 Altera® Agilex™ 5 E-series

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After launch of full featured Agilex™ 5 AXE5-Eagle Devkit in 2024, Arrow is excited to announce the second member of our Agilex™ 5 development kit line up the AXE5000 starter kit featuring first production device of Altera® Agilex™ 5 family.

The Altera® Agilex™ 5 series devices deliver excellent performance-per-watt in small form factors that make them ideal for midrange FPGA applications. Logic sizes range from 50k - 656k LEs and scale down to 15mm x 15mm through various package sizes. Agilex™ 5 is the industry's first FPGA infused with AI through the adaption of AI tensor blocks in the enhanced DSP block delivering high performance AI and DSP capabilities. In addition, the Agilex™ 5 offer achieves high performance due to second-generation Hyperflex™ FPGA Architecture. AXE5000 board has a HyperRAM memory, where this memory can be used for user data storage. The Agilex™ 5 device with the memory allows this kit to be used various applications including camera and vision applications. Configuration of the Agilex™ 5 FPGA is done via a 256 Mbit flash, which has additional free space for further data storage. An RGB LED, 3-axis accelerometer sensor, support starter labs and first design examples are included with this kit. Due to the low power consumption of the Agilex™ 5 E-series, the AXE5000 can be powered via USB, or via an additional 5V power supply.

Arrow's AXE5000 Starter Kit enables customers to evaluate the features of Intel Agilex™ 5 E-series and speed up development. With the CRUVI HS connector and Arduino MKR standard pads, various adapter boards can be connected to this kit to further create other applications. Arrow will provide a starter guide and demo designs for the AXE5000 on the Github. The board includes a full programmer and debugger for the FPGA. AXE5000 can also be modified and redesigned for customized versions, please contact Arrow for more details.

Partners:

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AXE5000 – Starter Kit

Key Features: Arrow AXE5000 Starter Kit

- Agilex™ 5 - A5EC008BM16AE6S (M16A)
  - * 85 kLE
  - * 116 DSP blocks
  - * 232 multipliers 18x19
  - * 3.05 Peak INT8 (TOPS)
  - * 4.47 Mb internal memory
  - * MIPI D-PHY interface
  - * LVDS pairs at 1.250 Mbps
  - * Secure Device Manager
- HyperRAM: 128 Mbit
- QSPI Config Flash: 256 Mbit
- 3-axis accelerometer
- RGB LED
- Push button and DIP switch
- CRUVI HS connector
- Arduino MKR standard pads (user IOs)
- On-board Programmer/Debugger
- 5V Power via USB
- Size: 25x71 mm²

Ordering number: AXE5000
Suggested Resale: 149 USD
Technical details: GitHub page